How does **Selection Sort** work? What are the steps?

7 5 2 9 1 1 4 3 6

Show the contents of the array above **each time** an **Selection Sort** changes it while sorting the array into **ascending order**.

How does **Insertion Sort** work? What are the steps?

7 5 2 9 1 1 4 3 6

Show the contents of the array above **each time** an **Insertion Sort** changes it while sorting the array into **ascending order**.

How does **Selection Sort** work? What are the steps?

7 5 2 9 1 1 4 3 6

Show the contents of the array above **each time** an **Selection Sort** changes it while sorting the array into **ascending order**.

Smallest → Final Position

7.	5	2	9	1.	1	4	3	6
1	5.	2	9	7	1.	4	3	6
✓	1	2✓	9	7	5	4	3	6
	✓	2	9.	7	5	4	3.	6
		✓	3	7.	5	4.	9	6
			✓	4	5✓	7	9	6
				✓	5	7.	9	6.
					✓	6	9.	7.
						✓	7	9
							✓	✓
✓ 1	1	2	3	4	5	6	7	9

• • = to be swapped

✓ = final or done!

How does **Insertion Sort** work? What are the steps?

7 5 2 9 1 1 4 3 6

Show the contents of the array above **each time** an **Insertion Sort** changes it while sorting the array into **ascending order**.

Sorted [Sublist]

[7]	5	2	9	1	1	4	3	6
[7.	5.]							
[5.	7	2.]						
[2	5	7	9]					
[2.	5	7	9	1.]				
[1	2.	5	7	9	1.]			
[1	1	2	5.	7	9	4.]		
[1	1	2	4.	5	7	9	3.]	
[1	1	2	3	4	5	7.	9	6.]
1	1	2	3	4	5	6	7	9 ✓

• • = to be sorted

✓ = done!

How does **Shell Sort** work? What are the steps?

8 5 3 2 2 9 3 7 4

Show the contents of the array above **each time** a **Shell Sort** changes it while sorting the array into **ascending order**.

8 5 3 2 2 9 3 7 4 6

Show the contents of the array above **each time** a **Shell Sort** changes it while sorting the array into **ascending order**.

9 5 6 4 3 9 3 8 2

Show the contents of the array above **each time** a **Shell Sort** changes it while sorting the array into **ascending order**.

How does **Shell Sort** work? What are the steps?

8 5 3 2 2 9 3 7 4 6 Almost Sorted n/2

Show the contents of the array above **each time** a **Shell Sort** changes it while sorting the array into **ascending order**.

											mid
8	5	3	2	2	9	3	7	4	6		10/2=5
8.					9.						
	3.					5.					
		3.					7.				
			2.					4.			
				2.					6.		
8	3	3	2	2	9	5	7	4	6		5/2=2
2.		3.		4.		5.		8.			
	2.		3.		6.		7.		9.		
[2]	2	3	3	4	6	5	7	8	9		2/2=1 *
[2	2]										
[2	2	3]									
[2	2	3	3]								
[2	2	3	3	4]							
[2	2	3	3	4	6]						
[2	2	3	3	4	6.	5.]					
[2	2	3	3	4	5	6	7]				
[2	2	3	3	4	5	6	7	8]			
[2	2	3	3	4	5	6	7	8	9]		
2	2	3	3	4	5	6	7	8	9		✓

8 5 3 2 2 9 3 7 4 Almost Sorted n/2

Show the contents of the array above **each time** a **Shell Sort** changes it while sorting the array into **ascending order**.

											mid
8	5	3	2	2	9	3	7	4			9/2=4
2.				4.				8.			
	5.				9.						
		3.				3.					
			2.				7.				
2	5	3	2	4	9	3	7	8			4/2=2
2.		3.		3.		4.		8.			
	2.		5.		7.		9.				
* [2]	2	3	5	3	7	4	9	8			2/2=1
[2	2]										
[2	2	3]									
[2	2	3	5]								
[2	2	3	5.	3.]							
[2	2	3	3	5	7]						
[2	2	3	3	5.	7	4.]					
[2	2	3	3	4	5	7	9]				
[2	2	3	3	4	5	7	9.	8.]			
✓ 2	2	3	3	4	5	7	8	9			

9 5 6 4 3 9 3 8 2 Almost Sorted n/2

Show the contents of the array above **each time** a **Shell Sort** changes it while sorting the array into **ascending order**.

											mid
9	5	6	4	3	9	3	8	2			9/2=4
2.				3.				9.			
	5.				9.						
		3.				6.					
			4.				8.				
2	5	3	4	3	9	6	8	9			4/2=2
2.		3.		3.		6.		9.			
	4.		5.		8.		9.				
[2]	4	3	5	3	8	6	9	9			2/2=1 *
[2	4]										
[2	4.	3.]									
[2	3	4	5]								
[2	3	4.	5	3.]							
[2	3	3	4	5	8]						
[2	3	3	4	5	8.	6.]					
[2	3	3	4	5	6	8	9]				
[2	3	3	4	5	6	8	9	9]			
2	3	3	4	5	6	8	9	9			✓

* = ordinary insertion sort

✓ = Done!